

Fusion draft: SST-17/40/42 Photon optics cluster

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About this fusion draft

This is an automatically assembled fusion draft: multiple SST manuscripts concatenated as chapters. It is intended for navigation and editing workflow, not as a final merged publication without manual cleanup.

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Chapter 1

SST-17 — Photon as torsion wave

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Electromagnetism as Propagating Torsion in a Hydrodynamic Vacuum:
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Abstract

We present a geometric reformulation of electromagnetism based on the hydrodynamics of a structured, director-bearing, inviscid and incompressible substrate. Using Cartan’s structure equations in a teleparallel (Weitzenböck) geometry, we identify the electromagnetic field strength 2-form as the *projection* of spacetime torsion along a fixed internal direction, $F \equiv \kappa u_a T^a$, which guarantees the Bianchi identity $dF = 0$. In this framework, the photon emerges not as a fundamental point particle, but as a massless *transverse shear-like* wave of the substrate’s director field. From an effective hydrodynamic (gauge-fixed) Lagrangian that is equivalent to the Maxwell action, we obtain the *linearized* vector wave equation and show that the wave speed is $c = \sqrt{\mathcal{K}/\rho_{\text{eff}}}$, i.e., the ratio of effective shear stiffness to effective mass density. Furthermore, we predict a vacuum magneto-optical response: in the presence of a parity-odd substrate coupling aligned with a background field, a *non-zero vacuum Verdet constant* (linear-in- B Faraday rotation) arises—distinct from the parity-even, quadratic-in- B birefringence of standard QED. This provides clean, falsifiable targets for high-intensity laser experiments and magnetar polarimetry.

1.1 Introduction

The quest to unify electromagnetism with the mechanical properties of space has a deep history in theoretical physics. While the standard model treats gauge fields as abstract entities defined on a static spacetime manifold, recent developments in Analogue Gravity [6] and Topological Fluid Dynamics [5] suggest that the vacuum may possess the characteristics of a *structured, superfluid-like, director-bearing* medium with an effective shear stiffness.

In this work, we explore the hypothesis that the vacuum behaves physically as an inviscid, incompressible fluid substrate characterized by a non-zero effective density ρ_{eff} and effective shear stiffness $\mathcal{K}$. Unlike early mechanical models which suffered from drag paradoxes, modern superfluid and nematic-fluid dynamics allow dissipationless flow and persistent topological structures. We utilize the language of Cartan geometry to map the kinematics of this substrate directly to the gauge fields of electrodynamics.

Specifically, we propose the following isomorphisms:

1. **Field as Projected Torsion:** The electromagnetic field tensor $F_{\mu\nu}$ is obtained by projecting the torsion 2-form along a fixed internal direction: $F_{\mu\nu} = \kappa u_a T^a_{\mu\nu}$, in a teleparallel background.
2. **The Photon as a Shear Wave:** Light is modeled as a propagating transverse oscillation of the substrate’s director field rather than a discrete particle.
3. **Vacuum Magneto-Optics:** A parity-odd coupling supplied by the substrate permits vacuum Faraday rotation (linear in B), providing a discriminator from QED’s parity-even B^2 birefringence.

This paper is organized as follows: Section II reviews the historical roots of hydrodynamic field theories. Section III establishes the geometric framework linking Cartan torsion to fluid defects. Section IV derives the photon wave equation from the hydrodynamic Lagrangian. Section V presents the derivation of the Vacuum Verdet constant and analyzes numerical bounds. Section VI outlines experimental tests in optical and superfluid systems.

1.2 Historical Context and Hydrodynamic Analogues

The concept of a structured medium underpinning physical phenomena is not new, but its modern formulation differs significantly from 19th-century approaches.

In 1858, Helmholtz [1] laid the foundations of vortex dynamics, proving that vortex lines in an inviscid fluid behave as conserved topological structures. Building on this, Lord Kelvin (1867) proposed the “Vortex Atom” hypothesis [2], suggesting that atoms were knotted vortices in a background fluid. While Kelvin’s specific model was abandoned due to the viscous drag problem in classical fluids, modern superfluid mechanics removes this objection. In superfluids (like He-II or Bose-Einstein Condensates), viscosity vanishes, and topological defects (vortices) persist indefinitely.

Contemporary research in Analogue Gravity has revived these ideas. Unruh [4] demonstrated that acoustic perturbations in moving fluids obey effective relativistic wave equations, giving rise to Lorentzian metrics dependent on the fluid flow. Volovik [5] extended this to quantum fluids, proposing that all gauge fields may arise from low-energy excitations of a condensed background. Our work formalizes this connection using differential geometry, treating the vacuum not merely as an analogue, but as a physical fluid substrate.

1.3 Geometric Framework: Cartan's Structure Equations

We describe the local geometry of the vacuum manifold $\mathcal{M}$ using Cartan's Method of Moving Frames. Let θ^i be the coframe 1-forms representing local displacements of the substrate, and ω^i_j be the connection 1-forms.

The two fundamental structure equations are:

$$\text{Torsion 2-form: } T^i = d\theta^i + \omega^i_j \wedge \theta^j, \quad (1.1)$$

$$\text{Curvature 2-form: } \mathbf{R}^i_j = d\omega^i_j + \omega^i_k \wedge \omega^k_j. \quad (1.2)$$

In our hydrodynamic model, we assume a Weitzenböck geometry where the connection vanishes globally ($\omega^i_j = 0$). This corresponds to a teleparallel medium where curvature vanishes but torsion is non-zero. In this limit,

$$T^i = d\theta^i. \quad (1.3)$$

1.3.1 Physical Identification (Minimal Projection)

Rather than equating the coframe directly with the $U(1)$ potential, we *project* the coframe-valued torsion onto a fixed internal direction u_a and define the electromagnetic field 2-form by

$$\boxed{F \equiv \kappa u_a T^a}. \quad (1.4)$$

With a covariantly constant u_a in teleparallel gauge, $dF = 0$ follows from $DT^a = 0$, ensuring the Bianchi identity and the existence of a potential A with $F = dA$. Thus, electromagnetism is interpreted as the dynamics of *propagating torsion projections*. Static torsion concentrations correspond to dislocations (charges/defects), while dynamic, propagating torsion corresponds to radiation.

1.4 The Photon as a Transverse Shear Wave

Standard Quantum Field Theory treats the photon as a point-like gauge boson. In this hydrodynamic model, we describe the photon as a delocalized, massless transverse oscillation (shear-like wave) of the substrate's director field. The director-bearing qualification is essential: a strictly incompressible inviscid fluid has zero shear modulus; here the substrate has an *effective* shear stiffness $\mathcal{K}$.

1.4.1 Kinematics of the Substrate

We define the local flow velocity $\mathbf{u}$ of the substrate in terms of the vector potential $\mathbf{A}(\mathbf{x}, t)$:

$$\mathbf{u} = \partial_t \mathbf{A}, \quad \nabla \cdot \mathbf{A} = 0. \quad (1.5)$$

The condition $\nabla \cdot \mathbf{A} = 0$ (Coulomb gauge) is interpreted as the incompressibility constraint $\nabla \cdot \mathbf{u} = 0$, suppressing longitudinal modes and leaving the observed transverse photon polarizations.

1.4.2 The Hydrodynamic Lagrangian

The dynamics of the substrate are governed by an effective Lagrangian density $\mathcal{L}$ constructed from kinetic energy and elastic energy:

$$\mathcal{L} = \frac{1}{2} \rho_{\text{eff}} \|\mathbf{u}\|^2 - \frac{1}{2} \mathcal{K} \|\nabla \times \mathbf{A}\|^2. \quad (1.6)$$

Identifying the speed of light as:

$$c = \sqrt{\frac{\mathcal{K}}{\rho_{\text{eff}}}}, \quad (1.7)$$

$[x=(0.7\text{cm},-0.3\text{cm}), y=(0.7\text{cm},0.3\text{cm}), z=(0\text{cm},1\text{cm}), \text{scale}=1.2]$
 $[->, \text{thick}, \text{black!60}] (-0.5,0,0) - (4+0.5,0,0) \text{ node[anchor=north west]} z \text{ (Propagation)}; [->, \text{thick}, \text{black!60}] (0,-1.2,0) - (0,1.2,0) \text{ node[anchor=south west]} y; [->, \text{thick}, \text{black!60}] (0,0,-1.2) - (0,0,1.2) \text{ node[anchor=south]} x;$
 $\text{in } 0,0.2,\dots,4$
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 $\sin(* 2.5* 180 / 3.14159));$
 $[<-, \text{thin}, \text{black}] (1.5, 0.8, 0.8) - (1.5, 1.5, 1.5) \text{ node[anchor=south]} \text{ Torsion Vector } \mathbf{T};$
 $[|-, \text{thin}, \text{red}] (0.6, -1.5, 0) - (3.1, -1.5, 0) \text{ node[midway, below]} \lambda;$
 $[\text{anchor=north}, \text{gray}, \text{align=center}] \text{ at } (3, -3, 0) \text{ Hydrodynamic Vacuum Substrate}$
 $(\rho_{\text{eff}}, \mathcal{K});$

Figure 1.1: **The Photon as a Propagating Torsional Shear Wave.** The electromagnetic field corresponds to a continuous transverse oscillation of the substrate's director field. The blue vectors trace a helical path, satisfying $\square \mathbf{A} = 0$.

this becomes the Coulomb-gauge form of the gauge-invariant Maxwell action $S = -\frac{1}{4\mu_0} \int F_{\mu\nu} F^{\mu\nu} d^4x$ under the identifications $\varepsilon_0 \equiv \rho_{\text{eff}}$ and $1/\mu_0 \equiv \mathcal{K}$.

1.4.3 Derivation of the Wave Equation

Applying the Euler–Lagrange equations to A_i :

$$\partial_t \left(\frac{\partial \mathcal{L}}{\partial (\partial_t A_i)} \right) = \rho_{\text{eff}} \partial_t^2 A_i, \quad (1.8)$$

$$\nabla \cdot \left(\frac{\partial \mathcal{L}}{\partial (\nabla A_i)} \right) = -\rho_{\text{eff}} c^2 [\nabla \times (\nabla \times \mathbf{A})]_i. \quad (1.9)$$

Combining these yields the vector wave equation:

$$\rho_{\text{eff}} (\partial_t^2 \mathbf{A} + c^2 \nabla \times \nabla \times \mathbf{A}) = 0. \quad (1.10)$$

Using the vector identity $\nabla \times \nabla \times \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$ and $\nabla \cdot \mathbf{A} = 0$, we recover the exact massless wave equation:

$$\boxed{\square \mathbf{A} = \left(\frac{1}{c^2} \partial_t^2 - \nabla^2 \right) \mathbf{A} = 0}. \quad (1.11)$$

This derivation demonstrates that Maxwell's equations for the photon emerge naturally from the mechanics of an incompressible, elastic fluid substrate, without postulating the field as a fundamental entity.

1.5 Prediction: Vacuum Faraday Rotation

A key consequence of this framework is that the vacuum behaves as a magneto-optical medium. In the presence of a parity-odd substrate coupling (background vorticity), the vacuum exhibits intrinsic birefringence.

1.5.1 Swirl-Induced Birefringence

Consider a photon (transverse shear wave) with frequency ω_γ and wavevector $\mathbf{k}$ propagating through a background fluid substrate rotating with vorticity $\boldsymbol{\Omega}$. In the rotating frame, the wave equation is modified by the Coriolis acceleration term $2\boldsymbol{\Omega} \times \partial_t \mathbf{A}$.

This leads to a modified dispersion relation for the Left (L) and Right (R) chiral modes:

$$\omega^2 \approx c^2 k^2 \pm 2\omega \mathbf{\Omega} \cdot \hat{k}. \quad (1.12)$$

Solving for the refractive index $n = ck/\omega$, we obtain the splitting:

$$n_{L/R} \approx 1 \pm \frac{\mathbf{\Omega} \cdot \hat{k}}{\omega_\gamma}. \quad (1.13)$$

This difference in phase velocity induces a rotation of the polarization plane (Faraday Rotation) by an angle $\Delta\theta$ over a propagation distance L :

$$\Delta\theta = \frac{\omega_\gamma L}{2c} (n_L - n_R) \approx \frac{L}{c} (\mathbf{\Omega} \cdot \hat{k}). \quad (1.14)$$

1.5.2 The Vacuum Verdet Constant

To formalize this in a gauge-invariant effective field theory, we introduce the Carroll–Field–Jackiw (CFJ) term:

$$S_{\text{odd}} = \frac{1}{2} \int p_\mu \epsilon^{\mu\nu\rho\sigma} A_\nu F_{\rho\sigma} d^4x, \quad (1.15)$$

where the axial vector p_μ is determined by the substrate background. Mapping the hydrodynamic vorticity to the magnetic field ($p_\parallel = \alpha_{\text{vac}} \mathbf{B}_0 \cdot \hat{k}$), we derive the **Vacuum Verdet Constant**:

$$\boxed{\mathcal{V}_{\text{vac}} \equiv \frac{\theta}{B L} \approx \frac{\alpha_{\text{vac}}}{2}}. \quad (1.16)$$

This predicts *linear-in- B* Faraday rotation in vacuum, distinct from the parity-even, quadratic-in- B birefringence ($\Delta n \propto B^2$) predicted by standard QED (Euler-Heisenberg).

1.5.3 Detectability and Regimes

The observation of this effect depends on overcoming the suppression by the microscopic scale r_c . Two promising regimes are:

1. **Extreme Path Lengths:** In cosmological or astrophysical baselines, the integrated rotation $\theta \sim \mathcal{V}_{\text{vac}} \int B_\parallel dl$ may become detectable. Crucially, this geometric rotation is **wavelength-independent**, distinguishing it from the plasma Rotation Measure (RM) which scales as λ^2 .
2. **Ultra-Strong Fields:** Environments like Magnetars ($B \sim 10^{10-11}$ T) and high-intensity laser setups (ELI-NP, LUXE) maximize $B_\parallel$. In these regimes, the linear scaling ($\theta \propto B$) of the hydrodynamic vacuum effect can be separated from the quadratic ($\theta \propto B^2$) QED effect.

Current laboratory constraints (e.g., PVLAS [10]) place upper bounds on α_{vac} . A non-null detection of linear vacuum rotation in future experiments would serve as direct evidence for the hydrodynamic substrate.

1.5.4 Comparison with QED

Standard Quantum Electrodynamics (QED) predicts a vacuum birefringence due to the Euler–Heisenberg Lagrangian:

$$\Delta n_{\text{QED}} \propto B^2. \quad (1.17)$$

Crucially, pure-vacuum QED is parity even: it yields *no Faraday rotation* (no linear-in- B term). In contrast, our hydrodynamic model predicts a birefringence that is **linear** in B (geometric). This distinct scaling behavior provides a clear “smoking gun” for experimental verification.

1.6 Experimental Implications

We propose several experimental avenues to test this hydrodynamic theory of light.

1.6.1 Optical Vortex Beams

High-order Laguerre–Gaussian (LG) beams carry orbital angular momentum (OAM). In our framework, high-OAM beams mimic the vorticity structure of the substrate. We predict anomalous group delays for these beams in rotating media or materials with strong spin–orbit coupling.

1.6.2 Superfluid Analogs

Bose–Einstein condensates and superfluid helium act as laboratory analogs. Rotating toroidal traps could display chiral splitting of phonon analogs, mirroring the substrate-induced parity-odd response.

1.6.3 Vacuum Birefringence Tests

Future high-intensity laser experiments (ELI-NP, LUXE) and observations of magnetars can probe vacuum optical properties. A deviation from the B^2 -only scaling would support the torsion-projection hypothesis.

1.7 Conclusion

We presented a geometric unification of electromagnetism and fluid dynamics. Identifying F as a projection of Cartan torsion in a teleparallel, director-bearing substrate recovers the Maxwell wave equation in a gauge-fixed hydrodynamic representation with $c = \sqrt{\mathcal{K}/\rho_{\text{eff}}}$.

A common objection to hydrodynamic vacuum models is the apparent violation of Lorentz invariance. However, as demonstrated in analogue gravity literature, **Lorentz invariance naturally emerges as an approximate acoustic symmetry** of the fluid equations in the low-energy limit [6]. Thus, this framework remains consistent with relativistic observations at standard energy scales.

Most importantly, this framework predicts a non-zero Vacuum Verdet constant ($\mathcal{V} \sim \alpha_{\text{vac}}/2$). While current experiments constrain this effect, it remains a falsifiable prediction distinguishable from the quadratic B^2 scaling of QED vacuum polarization. Future work will explore how topological defects in this substrate—knotted vortices—may give rise to massive fermions, aligning this framework with emerging vortex-string models of particle physics.

Author Contributions Statement

Omar Iskandarani: Conceptualization, Methodology, Formal Analysis, Investigation, Writing – Original Draft, and Writing – Review & Editing. The author confirms sole responsibility for the following study conception and design, data collection, analysis and interpretation of results, and manuscript preparation.

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Chapter 2

SST-40 — Photons and lasers (EN)

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Photons in Swirl-String Theory (SST):

Kinematics on Swirl Strings and Plotting a Laser Beam Omar Iskandarani

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Abstract

We model the photon as a one-dimensional, closed or open swirl string with phase $\phi(\mathbf{x}, t)$ propagating helically along the string. Spin (circular polarization) corresponds to the handedness of the local swirl clock; optical orbital angular momentum (OAM) with topological charge ℓ is the phase winding in the transverse plane. For laser beams we use the (paraxial) Gaussian beam and its Laguerre–Gaussian extension to plot intensity and phase fields. All formulas are SI-dimensional and calibrated to the SST scale $\Omega_0 = \|\mathbf{v}_0\|/r_c$. In the *Rosetta* mapping, the scalar phase mode plays the role of a Maxwell-like radiation sector with $\omega = ck$ in uniform backgrounds, while OAM corresponds to azimuthal phase winding.

2.1 Kinematics: photon as a helical mode on a swirl string

Let $\mathbf{X}(s, t)$ denote the string centerline with arclength parameter s and local tangent $\mathbf{t}$. A photon is modeled as a travelling phase wave on the string:

$$\phi(\mathbf{x}, t) = kz - \omega t + \ell \theta, \quad k = \frac{2\pi}{\lambda}.$$

where $k = 2\pi/\lambda$, $\omega = 2\pi f$, and (r, θ, z) are cylindrical coordinates along the propagation axis. *Spin*/polarization is the local swirl-clock handedness (left/right), and *OAM* is the integer winding $\ell \in \mathbb{Z}$ around the beam axis [2, 1].

SST clock and energy density. A convenient reference scale is

$$\Omega_0 = \frac{\|\mathbf{v}_\odot\|}{r_c} \quad [\text{s}^{-1}].$$

Dimensional check: $[\mathbf{v}_\odot] = \text{m s}^{-1}$, $[r_c] = \text{m}$, so Ω_0 is a frequency. Numerically this recovers the electron Compton scale in the canonical calibration and is used as a normalization point.

2.2 Energy, momentum, and polarization

For a single photon, $E = \hbar\omega$ and $p = \hbar k$ (standard field theory). Within SST, the energy is associated with an effective string line energy. Without committing to microstructure, the operative identification is

$$\boxed{E = \hbar\omega, \quad \mathbf{p} = \hbar\mathbf{k}, \quad \text{spin } S = \pm\hbar \leftrightarrow \text{swirl-clock left/right}}$$

where $+$ and $-$ correspond to left- and right-circular polarization, respectively.

2.3 Laser-beam model: Gaussian beam and LG modes

For a paraxial beam with waist w_0 at $z = 0$ ([1]):

$$w(z) = w_0 \sqrt{1 + (z/z_R)^2}, \quad z_R = \frac{\pi w_0^2}{\lambda}, \quad (2.1)$$

$$R(z) = z \left[1 + \left(\frac{z_R}{z} \right)^2 \right], \quad \zeta(z) = \arctan\left(\frac{z}{z_R} \right). \quad (2.2)$$

The TEM_{00} scalar field amplitude is

$$E_{00}(r, z) = E_0 \frac{w_0}{w(z)} \exp\left(-\frac{r^2}{w(z)^2} \right) \exp\left(ikz - i\omega t + i \frac{kr^2}{2R(z)} - i\zeta(z) \right),$$

The intensity is $I = \frac{1}{2}\epsilon_0 c |E|^2$ (units W m^{-2}). Laguerre–Gaussian (LG) with OAM ℓ and radial index p :

$$E_p^\ell(r, \theta, z) = E_{00} \left(\frac{\sqrt{2} r}{w(z)} \right)^{|\ell|} L_p^{|\ell|} \left(\frac{2r^2}{w(z)^2} \right) e^{i\ell\theta}.$$

For $\ell \neq 0$ there is an on-axis null; the ring maximum occurs at $r_{\max}(z) = w(z)\sqrt{|\ell|/2}$.

2.4 Numerical example and figures

Example parameters: $\lambda = 632.8 \text{ nm}$, $w_0 = 1.0 \text{ mm} \Rightarrow z_R = \pi w_0^2/\lambda = 4.9646 \text{ m}$ and beam divergence $\theta_{\text{div}} = \lambda/(\pi w_0) = 0.201 \text{ mrad}$.

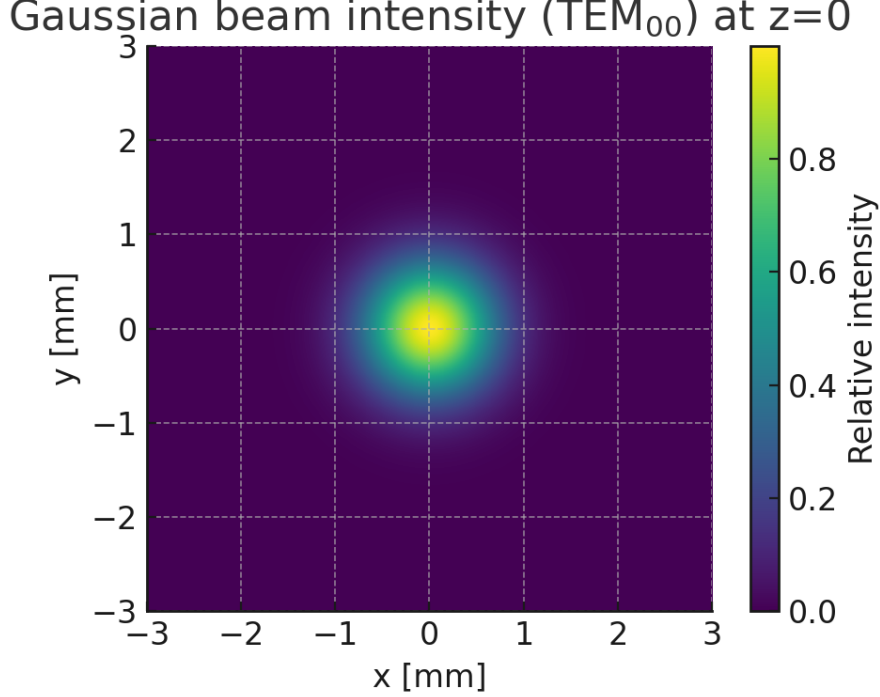


Figure 2.1: **Gaussian intensity (TEM₀₀) in the $z=0$ plane.** The map shows $I(r, 0) \propto \exp(-2r^2/w_0^2)$ with a central peak and radial Gaussian decay. With $\lambda = 632.8$ nm and $w_0 = 1.0$ mm this is the beam waist. The dimension of I is W m^{-2} ; relative scale shown.

Reading guide. Fig. 2.1 shows the TEM₀₀ intensity at the waist ($z = 0$). Fig. 2.2 shows the 2π phase winding for $\ell = 1$ (OAM, on-axis null). Fig. 2.3 confirms the known $w(z)$ scaling and the asymptotic divergence.

Known limits: $w(0) = w_0$, far-field $z \gg z_R$ gives opening angle $\theta_{\text{div}} = \lambda/(\pi w_0)$, and for $\ell \neq 0$ there is an axis null with ring maximum $r_{\text{max}}(z) = w(z)\sqrt{|\ell|/2}$ [2, 3].

2.5 Plotting recipe (algorithm)

1. Choose λ , w_0 ; compute $z_R = \pi w_0^2/\lambda$.
2. Define a 2D grid in the $z = 0$ plane; compute $I(r, 0) \propto e^{-2r^2/w_0^2}$ (TEM₀₀).
3. For OAM, take phase $\Phi = \ell \theta$ and (optionally) the LG envelope.
4. For a longitudinal cut: plot $w(z)$ and (optionally) $r_{\text{max}}(z)$.

2.6 Predictions (falsifiable) and edge cases

P1 (spin $\leftrightarrow$ swirl-clock). Polarization helicity coincides one-to-one with the handedness of the local swirl clock; spin-to-orbital conversion under strong focusing produces steps in ℓ (test via forked interferograms [3]). **P2 (OAM ring).** The on-axis null and ring radius r_{max} follow the LG scaling; deviations at extreme focusing (non-paraxial) predict measurable phase modulations. **Edge cases.** Non-paraxial ($w_0 \sim \lambda$), dispersive media, and the near field of structures (locally non-Gaussian) call for fully vectorial solutions.

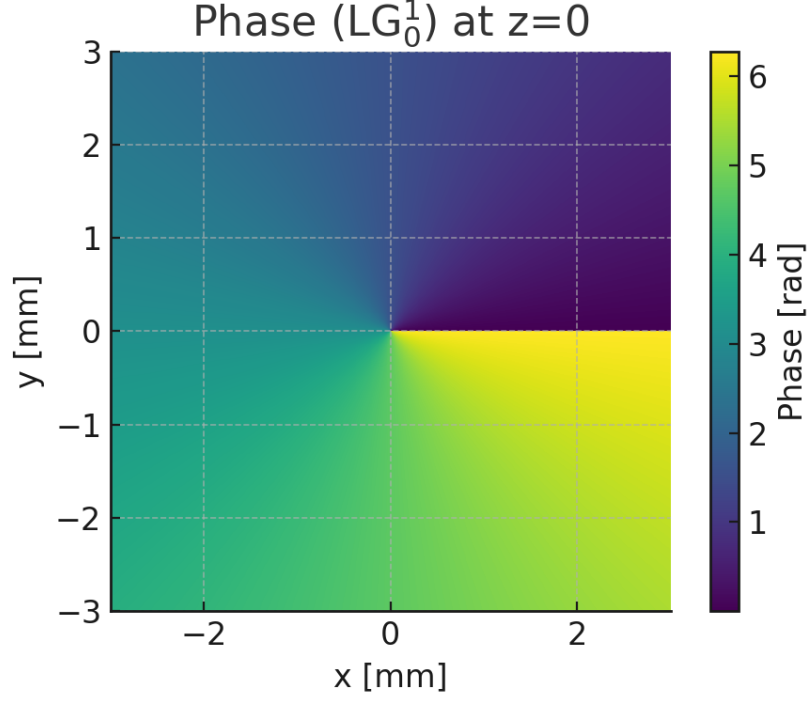


Figure 2.2: **Phase field for LG_0^1 with topological charge $\ell=1$ at $z=0$.** The phase $\Phi(\theta) = \ell \theta$ winds by 2π around the axis and has a singular core (dark “vortex”): zero intensity on-axis and a ring maximum at $r_{\text{max}}(0) = w_0/\sqrt{2} = 0.707$ mm. This visualizes optical OAM.

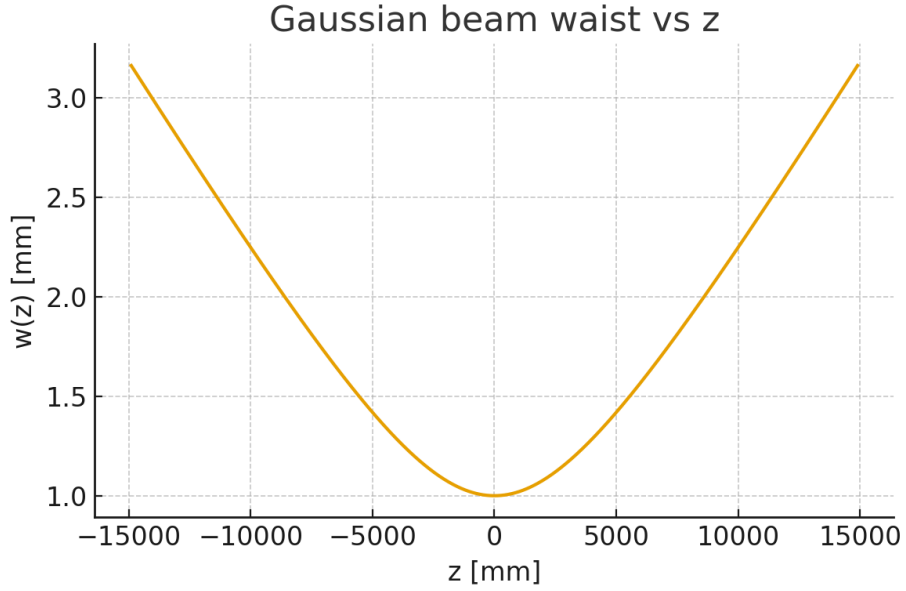


Figure 2.3: **Beam waist $w(z)$ versus z .** For $|z| \ll z_R$ the beam remains narrow; for $|z| \gg z_R$ one has $w(z) \approx |z|\theta_{\text{div}}$ with $\theta_{\text{div}} = \lambda/(\pi w_0)$. The Rayleigh range $z_R = 4.9646$ m marks the near-to far-field transition.

2.7 Mainstream mapping (Rosetta-style quick dictionary)

The following items summarize the translation to standard EM/QO language, consistent with Rosetta v0.6:

- **Scalar radiation mode:** phase field obeys $\partial_t^2\theta - c^2\nabla^2\theta = 0$ in uniform regions (luminal calibration).
- **Polarization:** spin $\pm\hbar$ maps to left/right circular polarization (handed swirl-clock).
- **OAM:** integer ℓ equals azimuthal phase winding; LG modes are standard paraxial solutions.
- **Energy/momentum:** $E = \hbar\omega$, $p = \hbar k$; intensity $I \propto |E|^2$ as in optics.
- **Analogue metric:** weak-field lensing/time-delay effects track gradients of the swirl energy fraction (see Rosetta).

Kid analogy. A photon is like a tiny corkscrew ripple that travels along an invisible string: turning left or right sets polarization; adding an extra twist per loop yields OAM rings.

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Chapter 3

SST-42 — Spiraling light

Transverse Orbital Angular Momentum and Off-Axis Tweezer Traps as a Maxwell-Limit Benchmark

Abstract

We provide a Swirl–String Theory (SST) interpretation of the spiraling light and optical Magnus-effect phenomenology developed by Spreeuw [1]. In standard electromagnetism, a tightly focused, linearly polarized beam excites a circular dipole whose radiation carries transverse orbital angular momentum (OAM), behaves as if emitted from a source displaced by $k^{-1} = \lambda/2\pi$, and produces spin-dependent off-axis equilibrium positions for atoms trapped in optical tweezers. In SST, photons are modeled as quantized swirl strings in an incompressible substrate; Maxwell electrodynamics appears as the long-wavelength ($\lambda \gg r_c$) limit of the swirl-field equations. We show that (i) the circular-dipole spiral field corresponds to an $\ell = \pm 1$ transverse swirl-string mode, (ii) the apparent source displacement is a geometric centroid of the swirl energy flux, and (iii) the off-axis tweezer displacement is a direct example of matter following gradients of the coarse-grained swirl energy density. The numerical size of possible SST corrections, scaling as $(kr_c)^2$, is shown to be $\mathcal{O}(10^{-16})$ for optical wavelengths, so that current experiments probe the pure Maxwell limit of SST while still providing a precise Rosetta case for photon topology and spin–orbit coupling.

3.1 Motivation

The analysis in Ref. [1] revisits the angular momentum content of tightly focused beams and the resulting forces on atoms. A key result is that a nominally linearly polarized beam, when focused to large numerical aperture, acquires a nontrivial spatial structure in both polarization and phase. This generates transverse spin, transverse OAM, and an effective source displacement of order k^{-1} , together producing an optical Magnus effect and spin-dependent off-axis trapping in optical tweezers.

Swirl-String Theory (SST) postulates that electromagnetic fields arise as collective excitations of quantized swirl strings in a flat, incompressible fluid-like substrate characterized by an effective density ρ_f , a characteristic swirl speed $\mathbf{v}_\zeta$, and a core length r_c . At wavelengths $\lambda \gg r_c$, the coarse-grained swirl dynamics reproduces Maxwell's equations. The spiraling light configuration is therefore an ideal Maxwell-limit benchmark: it is experimentally sharp, heavily constrained by angular-momentum conservation, and geometrically transparent.

In this note we map the main constructions of Ref. [1] into SST language and identify where potential SST-specific corrections could, in principle, enter.

3.2 Maxwell-Limit Summary of Spiraling Light

We briefly recall the core standard-electromagnetic results; detailed derivations are given in [1, 2, 3].

3.2.1 Circular dipole and transverse OAM

Consider an oscillating electric dipole

$$\mathbf{p}(t) = p_0(\hat{\mathbf{x}} \pm i\hat{\mathbf{z}})e^{-i\omega t}, \quad (3.1)$$

with angular frequency ω and wavenumber $k = \omega/c = 2\pi/\lambda$. The $\pm$ signs correspond to opposite senses of rotation. In the dipole plane ($y = 0$), the radiated field is (locally) linearly polarized but acquires a spiral phase dependence of the form $e^{\pm i\theta}$, where θ is the azimuthal angle in the xz -plane [1]. This phase winding corresponds to a transverse OAM of $\ell = \pm 1$ per photon for observers in the dipole plane, while on-axis observers see predominant spin.

The field behaves as if emitted by an effective source displaced by

$$\mathbf{R}_{\text{eff}} \simeq \pm \frac{1}{k} \hat{\mathbf{y}} = \pm \frac{\lambda}{2\pi} \hat{\mathbf{y}}, \quad (3.2)$$

a purely geometric offset set by the wavelength and the dipole rotation sense [1]. Equation (3.2) is dimensionally consistent, with k^{-1} having units of length.

3.2.2 Off-axis tweezer trapping

In an optical tweezer formed by a tightly focused beam, atoms experience a conservative potential $U(\mathbf{r}) \propto -\alpha I(\mathbf{r})$, where α is the scalar polarizability and I the local intensity. When the beam is configured to excite the circular dipole described above, the interference of the incident and re-radiated fields produces a transverse asymmetry of the intensity and Poynting vector. Ref. [1] shows that this leads to a stable equilibrium position displaced by $\mathcal{O}(\lambda)$ from the nominal optical axis, with opposite Zeeman sublevels trapped at opposite offsets.

Thus, the spiraling light construction provides:

- a tightly constrained mapping between dipole helicity, transverse OAM, and apparent source displacement;
- a specific, experimentally accessible force pattern on trapped atoms.

3.3 Swirl–String Photon Model in the Maxwell Limit

In SST, a single photon of frequency ω is modeled as a quantized swirl string: a closed filament of concentrated vorticity with core radius r_c and swirl speed $\mathbf{v}_\zeta$, embedded in an incompressible background of density ρ_f . The swirl energy density ρ_E is related to ρ_f and the local swirl speed, and the mass-equivalent density is $\rho_m = \rho_E/c^2$.

At long wavelengths ($\lambda \gg r_c$), the coarse-grained dynamics of these swirl strings reduces to linear wave equations for effective fields that can be identified with the electromagnetic field tensor $F_{\mu\nu}$. In this limit, SST predicts that:

1. the dispersion relation is $\omega = ck$ up to corrections of order $(kr_c)^2$;
2. the field carries total angular momentum $\mathbf{J} = \mathbf{L} + \mathbf{S}$, with $L_z = \ell\hbar$ and $S_z = \sigma\hbar$ emerging from the topology and internal twist of the swirl string;
3. the leading corrections to Maxwell electrodynamics are suppressed by a dimensionless factor

$$\epsilon(k) \sim (kr_c)^2 = \left(\frac{2\pi r_c}{\lambda}\right)^2. \quad (3.3)$$

Numerically, taking $r_c = 1.40897017 \times 10^{-15}$ m and a representative optical wavelength $\lambda = 1.0 \times 10^{-6}$ m, we obtain

$$\epsilon(k) = \left(\frac{2\pi r_c}{\lambda}\right)^2 \approx 7.8 \times 10^{-17}, \quad (3.4)$$

a dimensionless correction that is far below the sensitivity of current optical experiments. Thus the spiraling light configurations studied in [1] lie deep inside the Maxwell regime of SST.

3.4 SST Translation of Spiraling Light

3.4.1 Circular dipole as a localized swirl-string source

In the Maxwell description, the circular dipole is specified by $\mathbf{p}(t)$ and produces a complex field $\mathbf{E}(\mathbf{r}, t)$ with spiral phase. In SST, we instead specify a localized swirl-string source whose internal degree of freedom (the SwirlClock S_t^ζ) precesses with frequency ω and helicity $\sigma = \pm 1$. The far-field, after coarse-graining, is described by an effective complex scalar phase

$$\Phi_{\text{SST}}(\mathbf{r}, t) = kz - \omega t + \ell\theta, \quad (3.5)$$

with $\ell = \pm 1$ encoding the transverse phase winding inherited from the swirl-string topology. The emergent electromagnetic field is then reconstructed from gradients and time-derivatives of Φ_{SST} , reproducing the same $e^{\pm i\theta}$ phase structure found in [1].

In this mapping:

- the internal swirl-string twist maps to photon spin σ ;
- the spatial winding of the swirl string in the transverse plane maps to photon OAM ℓ ;
- the dipole plane corresponds to an observation slice where OAM dominates over spin in the measured angular-momentum density, consistent with the transverse OAM interpretation.

3.4.2 Effective source displacement as a swirl energy centroid

Equation (3.2) can be rephrased as a statement about the intensity-weighted centroid of the field in the dipole plane. Let $I(\mathbf{r}_\perp)$ be the time-averaged Poynting flux magnitude through a transverse plane, with $\mathbf{r}_\perp = (x, z)$ and y the transverse offset. Define the effective emission point as

$$\mathbf{R}_{\text{eff}} = \frac{\int \mathbf{r}_\perp I(\mathbf{r}_\perp) d^2 r_\perp}{\int I(\mathbf{r}_\perp) d^2 r_\perp}. \quad (3.6)$$

In the Maxwell analysis of [1], the special structure of $I(\mathbf{r}_\perp)$ for the circular dipole leads directly to $\mathbf{R}_{\text{eff}} \simeq \pm k^{-1} \hat{\mathbf{y}}$.

In SST, $I(\mathbf{r}_\perp)$ is proportional to the swirl energy flux, derived from $\rho_E(\mathbf{r})$ and the local swirl velocity. In the $\lambda \gg r_c$ regime, the proportionality is linear and isotropic, so that the centroid definition (3.6) is unchanged. Consequently, SST predicts the same universal offset (3.2) in the Maxwell limit, with any deviation only entering through the tiny $(kr_c)^2$ corrections of Eq. (3.3).

3.4.3 Off-axis tweezer displacement as motion in a swirl potential

The optical tweezer potential $U(\mathbf{r}) \propto -\alpha I(\mathbf{r})$ can be interpreted in SST as a potential energy landscape generated by the swirl energy density $\rho_E(\mathbf{r})$ and its coupling to the internal degrees of freedom of the trapped atom. The force is

$$\mathbf{F}(\mathbf{r}) = -\nabla U(\mathbf{r}) \propto \nabla I(\mathbf{r}), \quad (3.7)$$

identical in form to the Maxwell description. The spin-dependent displacement arises because the circular dipole component—and therefore the local swirl-string mode that is most strongly excited—changes with the internal state of the atom. Different Zeeman sublevels couple to different helicities σ , thus sampling slightly different swirl energy landscapes.

In SST terms:

1. the internal two-level system of the atom is treated as a localized swirl-clock degree of freedom S_t^σ ,
2. the focused beam prepares a structured swirl-string field with nontrivial ℓ and σ ,
3. the equilibrium positions are minima of the effective potential defined by the swirl energy density around the atom.

Because the underlying field is in the Maxwell limit, the resulting equilibrium displacements are of order λ and match those computed in [1]. SST neither modifies nor conflicts with these predictions at currently accessible optical parameters.

3.5 Outlook and SST-Specific Opportunities

The spiraling light construction offers a clean Rosetta example for SST:

- It provides a concrete mapping between swirl-string topology (ℓ, σ) and measurable spin-orbit effects in focused beams.
- It realizes a simple example of matter moving in a structured swirl energy landscape, with spin-dependent equilibrium positions.
- It probes the regime $\lambda \gg r_c$, where SST must reproduce Maxwell's equations to extremely high precision.

A natural next step is to consider regimes where the beam waist approaches microscopic scales or where multiple swirl-string modes interfere in strongly nonparaxial geometries. Within SST, higher-derivative corrections suppressed by $(kr_c)^2$ could then, in principle, generate small deviations from the pure Maxwell predictions in angular momentum flow or equilibrium positions. For $\lambda \sim 1\text{ }\mu\text{m}$ and $r_c \sim 10^{-15}\text{ m}$, the estimated correction $\epsilon(k) \sim 8 \times 10^{-17}$ implies that such effects are currently unobservable, but they offer a well-defined target for future high-precision tests.

In summary, the spiraling light and optical Magnus-effect phenomenology of Ref. [1] is fully compatible with the SST Canon. It provides a useful benchmark problem for the SST photon model, a laboratory example of matter–swirl coupling in the Maxwell limit, and a potential springboard for constraining SST corrections beyond Maxwell electrodynamics.

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